Assignment 3

Write a Java program to implement method overloading and method overriding in Java.

1.) Write a class Shapes. Find the area of basic shapes using Polymorphism

2.) Write a Java program to create a parent class Hill stations and three subclasses by hill station names. Subclasses extend the superclass and override methods famosis food () and famousfor (). Call the methods famousfood () and famousfor () by parent class. It should refer to the base class object and base class method overrides superclass method, baseclass method is invoked at runtime.

OBJECTIVE :

1.) To study method overloading
2.) To study method overriding in Java.

THEORY :

1.) Method overloading & it's advantages

- Method overloading is a feature in object oriented programming where multiple methods

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in same class have same name but diff parameters (type, no. or order)

— Advantages :

- Improves code redability
- Provides flexiblity in method usage
- Supports compile-time polymorphism.
- Reduces need for different method names.

2.) Method overriding and it's rules :

— Method overriding occurs when a subclass provides a specific implementation of a method that is already ~~provid~~ defined in it's superclass.

— Rules :

- Method name must be same as parent class.
- Parameters must be identical.
- Must follow inheritance.
- Cannot reduce access level (eg. public → private not allowed).
- Return type must be same or covariant.

CONCLUSION :

Thus, we have successfully implemented method overloading and method overriding in Java.

FAQs :

Q.1) Can we overload java

1 Yes, we can

FAQs :

Q1.) Can we overload java main() method ?
- Yes, we can overload the main() method in Java by changing it's parameter list.
- However, only public static void main (String[] args) is used by JVM as the entry point.

Q2.) Can we declare the overloaded methods as final ?
- Yes, the overloaded methods can be declared as final.
- final only prevents ~~or~~ overriding, not overloading.

Q3.) Can overloaded method be overridden.
- No, overloading and overriding are different concepts.
- However, a method that is overloaded in a superclass can be overridden in a subclass if the signature matches.

Q4.) What is method overriding ?
- Method overriding is when a subclass provides a specific implementation of a method already defined in it's superclass, ~~ut~~ with the same name and parameters.

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Q5.) What happens if we change the arguments of overriding method?

- If we change the arguments, of an overriding method, it is no longer overriding.
- It becomes method overloading.

Q6.) Can we change the return type of overriding method from no. to integer type?

- Yes, it is allowed if it is a covariant return type.
- Since integer is a subclass of Number, changing return type from number to integer is valid in overriding.